

# An Entropy–Transport Attack on Frankl’s Union-Closed Sets Conjecture

Open Conjecture Formalizations

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## Abstract

This note develops a possible route toward Frankl’s union-closed sets conjecture by trying to combine two proof paradigms that have historically stalled separately: entropy growth under union, and pairing/matching arguments. The organizing idea is that the entropy method fails precisely when the union map has large collision/dependence cost, while pairing methods fail precisely when attempted injections collide. We propose treating these as the same obstruction. Either a well-chosen coupling of two random members of the family produces an entropy contradiction, or the dual obstruction to such a coupling should produce a Hall-type certificate rigid enough to force a Frankl element.

## 1 The Basic Setup

Let  $\mathcal{F}$  be a finite union-closed family of finite sets. After replacing the ambient set by  $\bigcup_{A \in \mathcal{F}} A$ , write the ground set as  $[n]$ . Equivalently,

$$\mathcal{F} \subseteq \{0, 1\}^n$$

is closed under coordinatewise OR.

Let  $X$  be uniformly distributed on  $\mathcal{F}$ , and write

$$p_i = \mathbb{P}[X_i = 1].$$

Frankl’s conjecture is:

$$\max_i p_i \geq \frac{1}{2},$$

provided  $\mathcal{F}$  has at least one nonempty member.

**Conjecture 1** (Frankl). *If  $\mathcal{F}$  is finite, nontrivial, and union-closed, then some element of the ground set belongs to at least half of the members of  $\mathcal{F}$ .*

For this note, assume toward contradiction that  $\mathcal{F}$  is a counterexample:

$$0 < p_i < \frac{1}{2} \quad \text{for every active coordinate } i.$$

## 2 The Join-Semilattice View

The family  $\mathcal{F}$  is a finite join-semilattice with join operation

$$A \vee B = A \cup B.$$

Each coordinate  $i$  gives a join-homomorphism

$$\chi_i : \mathcal{F} \rightarrow \mathbf{2}, \quad \chi_i(A) = 1 \iff i \in A,$$

because

$$\chi_i(A \vee B) = \chi_i(A) \vee \chi_i(B).$$

The maps  $\chi_i$  jointly separate points if the ground set has been reduced by identifying indistinguishable elements.

Thus Frankl can be read as a sharp fiber-size statement:

$$\text{one separating join-homomorphism } \chi_i : \mathcal{F} \rightarrow \mathbf{2} \text{ has } |\chi_i^{-1}(1)| \geq |\mathcal{F}|/2.$$

The categorical viewpoint is useful because it isolates the structure:

$$\text{finite join-semilattice} \quad + \quad \text{separating coordinates to } \mathbf{2} \quad + \quad \text{uniform counting measure}.$$

The hard part is the last phrase. Pure structure-preserving maps do not, by themselves, remember the sharp counting threshold.

## 3 Known Quantitative Barriers

### 3.1 The iid Entropy Barrier

Let  $X, Y$  be independent uniform samples from  $\mathcal{F}$ , and set

$$Z = X \vee Y.$$

Since  $\mathcal{F}$  is union-closed,  $Z$  is supported on  $\mathcal{F}$ , so

$$H(Z) \leq \log |\mathcal{F}| = H(X).$$

On coordinate  $i$ ,

$$q_i := \mathbb{P}[Z_i = 1] = 2p_i - p_i^2.$$

The coordinate entropy gain is

$$h(q_i) - h(p_i),$$

where  $h(t) = -t \log t - (1-t) \log(1-t)$ .

This gain is positive as long as

$$p_i < 2p_i - p_i^2 < 1 - p_i,$$

which gives

$$p_i < \frac{3 - \sqrt{5}}{2}.$$

This is the natural barrier faced by iid OR-sampling: above that threshold, OR pushes the coordinate past the entropy peak at  $1/2$ .

### 3.2 Average-Size Bounds Lose Dimension

Average-size methods estimate

$$\sum_i p_i = \mathbb{E}[|X|].$$

But Frankl requires one  $p_i \geq 1/2$ . Any lower bound on the sum loses force when the effective number of coordinates is large.

### 3.3 Pairing Arguments Collide

If  $\{i\} \in \mathcal{F}$ , then the map

$$A \mapsto A \cup \{i\}$$

injects members omitting  $i$  into members containing  $i$ , proving that  $i$  appears in at least half the family.

For a larger set  $S \in \mathcal{F}$ , the natural map

$$A \mapsto A \cup S$$

can collapse many different  $A$  to the same image. This is the basic local failure of pairing proofs.

## 4 Coupled OR Transport

The iid choice of  $(X, Y)$  is only one coupling of the uniform distribution on  $\mathcal{F}$  with itself. Let  $\Pi(\mathcal{F})$  be the set of probability measures  $\pi$  on  $\mathcal{F} \times \mathcal{F}$  whose two marginals are both uniform.

For  $\pi \in \Pi(\mathcal{F})$ , let  $(X, Y) \sim \pi$  and

$$Z = X \vee Y.$$

Define

$$q_i(\pi) = \mathbb{P}_\pi[Z_i = 1].$$

Since  $Z$  is supported on  $\mathcal{F}$ ,

$$H(Z) \leq H(X) = \log |\mathcal{F}|.$$

Now decompose entropy into coordinate entropy minus total correlation:

$$H(W) = \sum_i H(W_i) - \text{TC}(W),$$

where

$$\text{TC}(W) = \sum_i H(W_i) - H(W).$$

Thus

$$H(Z) - H(X) = \sum_i (h(q_i(\pi)) - h(p_i)) - (\text{TC}(Z) - \text{TC}(X)).$$

**Definition 1** (Entropy advantage). *For a coupling  $\pi \in \Pi(\mathcal{F})$ , define*

$$\Phi(\pi) = \sum_i (h(q_i(\pi)) - h(p_i)) - (\text{TC}(Z) - \text{TC}(X)).$$

Since  $Z$  is supported on  $\mathcal{F}$ , every coupling satisfies

$$\Phi(\pi) = H(Z) - H(X) \leq 0.$$

Therefore:

**Proposition 1** (Entropy contradiction criterion). *If, under the assumption  $p_i < 1/2$  for all  $i$ , there exists  $\pi \in \Pi(\mathcal{F})$  with  $\Phi(\pi) > 0$ , then  $\mathcal{F}$  is not a counterexample.*

The difficulty is to choose  $\pi$ . The iid coupling often makes  $q_i(\pi) = 2p_i - p_i^2$  too large for coordinates near  $1/2$ . The diagonal coupling makes  $q_i(\pi) = p_i$  and gives no coordinate gain. We want a positively correlated but not fully diagonal coupling:

$$p_i < q_i(\pi) < 1 - p_i \quad \text{for all } i.$$

Equivalently, since

$$q_i(\pi) = 2p_i - r_i(\pi), \quad r_i(\pi) = \mathbb{P}_\pi[X_i = Y_i = 1],$$

we want

$$3p_i - 1 < r_i(\pi) < p_i.$$

For coordinates close to  $1/2$ , this asks for very strong positive correlation, but still leaves room below the diagonal.

## 5 The Linear Feasibility Problem and Its Dual

Fix a set of coordinates  $I$  and desired targets  $\tau_i$  for  $i \in I$ . For example,  $\tau_i = 1/2$  asks to center coordinate  $i$  at the entropy peak. Let

$$c_i(A, B) = \mathbf{1}_{i \in A \cup B}.$$

The centering problem is the following transportation feasibility problem:

$$\begin{aligned} \text{find } & \pi_{A,B} \geq 0 & (A, B \in \mathcal{F}) \\ \text{such that } & \sum_B \pi_{A,B} = 1/m & (A \in \mathcal{F}), \\ & \sum_A \pi_{A,B} = 1/m & (B \in \mathcal{F}), \\ & \sum_{A,B} c_i(A, B) \pi_{A,B} = \tau_i & (i \in I), \end{aligned}$$

where  $m = |\mathcal{F}|$ .

By Farkas' lemma, this system is infeasible if and only if there exist row potentials  $u_A$ , column potentials  $v_B$ , and coordinate weights  $\lambda_i$  such that

$$u_A + v_B + \sum_{i \in I} \lambda_i c_i(A, B) \geq 0 \quad \text{for all } A, B \in \mathcal{F},$$

but

$$\frac{1}{m} \sum_A u_A + \frac{1}{m} \sum_B v_B + \sum_{i \in I} \lambda_i \tau_i < 0.$$

Thus any failure to center a collection of coordinates has a very concrete dual certificate: a weighted inequality on the OR-incidence matrix that is nonnegative on every pair  $(A, B)$  but negative on the demanded marginals.

For the window problem one replaces the equality constraints by

$$p_i + \gamma \leq \sum_{A, B} c_i(A, B) \pi_{A, B} \leq 1 - p_i - \gamma$$

and maximizes  $\gamma$ . The dual then has two nonnegative coordinate weights for each  $i$ , one for the lower wall and one for the upper wall. Boundary examples with  $p_i = 1/2$  are already explained by this dual: the diagonal coupling gives  $q_i = p_i$ , while the upper wall is also  $1 - p_i = p_i$ , so no positive margin is possible.

## 6 One Coordinate Can Be Understood Exactly

The first sanity check is the one-coordinate problem. It does not use union-closure at all.

**Proposition 2** (Single-coordinate OR range). *Fix a coordinate  $i$ , and let*

$$R_i = \{A \in \mathcal{F} : i \in A\}, \quad L_i = \{A \in \mathcal{F} : i \notin A\}.$$

*Write  $r = |R_i|$ ,  $\ell = |L_i|$ ,  $m = r + \ell$ , and  $p_i = r/m$ . As  $\pi$  ranges over uniform couplings of two uniform samples from  $\mathcal{F}$ , the possible values of*

$$q_i(\pi) = \mathbb{P}_\pi[i \in X \cup Y]$$

*fill the interval*

$$p_i \leq q_i(\pi) \leq \min(1, 2p_i).$$

*Proof.* By the Birkhoff-von Neumann theorem, it is enough to consider permutation couplings and then take convex combinations. For a permutation coupling  $Y = \sigma(X)$ , every row  $A \in R_i$  contributes to  $q_i$ . In addition, a row  $A \in L_i$  contributes precisely when  $\sigma(A) \in R_i$ . If  $t$  denotes the number of rows of  $L_i$  mapped into columns of  $R_i$ , then

$$q_i = \frac{r + t}{m}.$$

The possible values of  $t$  range from 0 to  $\min(\ell, r)$ : assign as many columns of  $R_i$  as desired to rows of  $L_i$ , and fill the remaining rows and columns bijectively. Hence the permutation values range from  $r/m$  to  $(r + \min(\ell, r))/m = \min(1, 2r/m)$ , and convex combinations fill the interval.  $\square$

Consequently, for a strictly light coordinate  $0 < p_i < 1/2$  the window

$$p_i < q_i < 1 - p_i$$

is always possible for that coordinate alone. However, exact centering  $q_i = 1/2$  is possible only when

$$p_i \geq \frac{1}{4}.$$

This corrects the naive “center every light coordinate” guess: very rare coordinates can be moved upward, but not all the way to the entropy peak in a single OR step.

## 7 The Assignment-Dual Form of Critical Centering

The computational evidence below suggests that critical light coordinates can often be centered simultaneously. It is useful to put that claim into a form that has no hidden analysis.

Fix a coordinate set  $I$ . For a permutation  $\sigma$  of  $\mathcal{F}$ , write

$$q(\sigma)_i = \frac{1}{m} |\{A \in \mathcal{F} : i \in A \cup \sigma(A)\}| \quad (i \in I).$$

By Birkhoff-von Neumann, the possible vectors  $q_I(\pi)$  over all uniform self-couplings are exactly the convex hull of the permutation vectors  $q(\sigma)$ .

**Proposition 3** (Assignment dual for critical centering). *The center vector  $(1/2, \dots, 1/2) \in \mathbb{R}^I$  belongs to this convex hull if and only if, for every weight vector  $\lambda \in \mathbb{R}^I$ ,*

$$\max_{\sigma \in S_{\mathcal{F}}} \frac{1}{m} \sum_{A \in \mathcal{F}} \sum_{i \in I} \lambda_i \mathbf{1}_{i \in A \cup \sigma(A)} \geq \frac{1}{2} \sum_{i \in I} \lambda_i.$$

*Proof.* This is the separating-hyperplane theorem applied to the finite polytope  $\text{conv}\{q(\sigma) : \sigma \in S_{\mathcal{F}}\}$ . If the center is outside the polytope, some linear functional  $\lambda$  is larger at the center than on every permutation vector. Conversely, such a  $\lambda$  is exactly a separating functional.  $\square$

Thus critical centering is equivalent to a weighted assignment problem. For fixed  $\lambda$ , define

$$w_{\lambda}(A, B) = \sum_{i \in I} \lambda_i \mathbf{1}_{i \in A \cup B}.$$

The desired inequality is

$$\max_{\sigma} \sum_A w_{\lambda}(A, \sigma(A)) \geq \frac{m}{2} \sum_i \lambda_i.$$

The weight matrix has special join form:

$$w_{\lambda}(A, B) = \sum_{i \in A \cap I} \lambda_i + \sum_{i \in I \setminus A} \lambda_i \mathbf{1}_{i \in B}.$$

So the row contribution already present in  $A$  is fixed; only the missing positive or negative coordinate weights are assigned to columns.

This exposes a thinner quantitative barrier. The diagonal permutation gives  $q_i = p_i$ , and therefore proves the dual inequality whenever all coordinate weights are nonpositive, since  $\lambda_i p_i > \lambda_i/2$  when  $\lambda_i < 0$  and  $p_i < 1/2$ . The average over all permutations gives

$$\mathbb{E}_{\sigma} q(\sigma)_i = 2p_i - p_i^2,$$

which is at least  $1/2$  when

$$p_i \geq 1 - \frac{1}{\sqrt{2}} \approx 0.2929.$$

Therefore nonnegative weights on the upper part of the critical range are handled by the random-permutation average. The remaining genuinely hard zone is narrow but important:

$$\frac{1}{4} \leq p_i < 1 - \frac{1}{\sqrt{2}},$$

especially when these rare positive coordinates must be centered while negative coordinates are kept from being pushed upward. Thus the real assignment problem is a signed compromise between diagonal behavior and outward-pushing behavior. A proof of the assignment dual would need to show that the maximizing permutation beats the random average in precisely this rare-coordinate regime.

**Problem 1** (Weighted critical assignment inequality). *Let  $\mathcal{F}$  be union-closed, let  $I = \{i : 1/4 \leq p_i < 1/2\}$ , and let  $\lambda \in \mathbb{R}^I$ . Prove*

$$\max_{\sigma \in S_{\mathcal{F}}} \frac{1}{m} \sum_A w_{\lambda}(A, \sigma(A)) \geq \frac{1}{2} \sum_i \lambda_i,$$

*or classify exactly which signed weights can fail. Any genuine failure gives a concrete separating certificate for the critical-centering conjecture.*

## 7.1 The Assignment LP Dual

For a fixed  $\lambda$ , write

$$s_{\lambda}(A) = \sum_{i \in A \cap I} \lambda_i, \quad g_{\lambda}(A, B) = \sum_{i \in I \setminus A} \lambda_i \mathbf{1}_{i \in B}.$$

Then

$$w_{\lambda}(A, B) = s_{\lambda}(A) + g_{\lambda}(A, B).$$

The row term is independent of the assignment:

$$\sum_{A \in \mathcal{F}} s_{\lambda}(A) = m \sum_{i \in I} \lambda_i p_i.$$

Therefore the weighted critical assignment inequality is equivalent to

$$\max_{\sigma} \sum_A g_{\lambda}(A, \sigma(A)) \geq m \sum_{i \in I} \lambda_i \left( \frac{1}{2} - p_i \right).$$

Set

$$d_i = \frac{1}{2} - p_i > 0.$$

The normalized problem is:

$$\max_{\sigma} \sum_A \sum_{i \in I \setminus A} \lambda_i \mathbf{1}_{i \in \sigma(A)} \geq m \sum_i \lambda_i d_i.$$

This form is conceptually useful. The positive part of  $\lambda$  asks rows missing  $i$  to be matched into columns containing  $i$ . The negative part asks rows missing  $i$  to avoid such columns. Since the right side is also signed, large negative weights are usually harmless; they make the demanded lower bound easier, and the diagonal assignment realizes zero gain on every missing coordinate.

Now pass to the bipartite assignment LP:

$$\begin{aligned} & \text{maximize} && \sum_{A, B} g_{\lambda}(A, B) x_{A, B} \\ & \text{subject to} && \sum_B x_{A, B} = 1 && (A \in \mathcal{F}), \\ & && \sum_A x_{A, B} = 1 && (B \in \mathcal{F}), \\ & && x_{A, B} \geq 0. \end{aligned}$$

The Birkhoff-von Neumann theorem says its optimum is attained by a permutation. Its dual is

$$\begin{aligned} & \text{minimize} \quad \sum_A \alpha_A + \sum_B \beta_B \\ & \text{subject to} \quad \alpha_A + \beta_B \geq g_\lambda(A, B) \quad (A, B \in \mathcal{F}). \end{aligned}$$

Thus failure of critical centering is equivalent to the existence of  $\lambda$ , row potentials  $\alpha_A$ , and column potentials  $\beta_B$  such that

$$\alpha_A + \beta_B \geq \sum_{i \in I \setminus A} \lambda_i \mathbf{1}_{i \in B} \quad (A, B \in \mathcal{F}),$$

but

$$\sum_A \alpha_A + \sum_B \beta_B < m \sum_i \lambda_i d_i.$$

This is the finite certificate we should try to rule out.

**Problem 2** (Potential obstruction). *Classify triples  $(\lambda, \alpha, \beta)$  satisfying the dual inequalities with objective below  $m \sum_i \lambda_i d_i$ . Show that, after quotienting clones and deleting dominated critical coordinates, no such triple exists.*

## 7.2 The Remaining Obstruction Cone

Two elementary assignments already rule out most possible signs.

First, the diagonal assignment  $\sigma(A) = A$  gives

$$\sum_A g_\lambda(A, A) = 0.$$

Hence any failing  $\lambda$  must satisfy

$$\sum_i \lambda_i d_i > 0.$$

Second, averaging over a uniformly random permutation gives

$$\mathbb{E}_\sigma \sum_A g_\lambda(A, \sigma(A)) = m \sum_i \lambda_i p_i (1 - p_i).$$

Therefore any failing  $\lambda$  must also satisfy

$$\sum_i \lambda_i p_i (1 - p_i) < \sum_i \lambda_i d_i.$$

Equivalently, with

$$e_i = p_i(1 - p_i) - d_i = 2p_i - p_i^2 - \frac{1}{2},$$

a failing  $\lambda$  must lie in the cone

$$\sum_i \lambda_i d_i > 0, \quad \sum_i \lambda_i e_i < 0.$$

For  $p_i \geq 1 - 1/\sqrt{2}$ , one has  $e_i \geq 0$ . Thus positive mass on high-critical coordinates helps the random-permutation bound rather than hurting it. The only positive weights that can drive the second inequality must concentrate on the narrow low-critical range

$$\frac{1}{4} \leq p_i < 1 - \frac{1}{\sqrt{2}}.$$

**Lemma 1** (Necessary cone for a dual obstruction). *If  $\lambda$  violates the weighted critical assignment inequality, then*

$$\sum_i \lambda_i \left( \frac{1}{2} - p_i \right) > 0 \quad \text{and} \quad \sum_i \lambda_i \left( 2p_i - p_i^2 - \frac{1}{2} \right) < 0.$$

*Proof.* The first inequality follows because the diagonal assignment gives left-hand side 0 in the normalized problem. The second follows because the optimum is at least the average value of the normalized objective over all permutations.  $\square$

This cone is a useful target for both proof and search. In particular, a face-stripped obstruction cannot be all-negative, cannot be supported only on high-critical positive coordinates, and cannot be explained by coordinate order. It must be a genuinely signed tradeoff between rare positive coordinates and compensating negative coordinates.

### 7.3 Single-Positive Directions

The one-coordinate part of the remaining cone is completely harmless.

**Proposition 4** (One-coordinate dual inequality). *Let  $I = \{i\}$  and  $1/4 \leq p_i < 1/2$ . Then the weighted critical assignment inequality holds for every  $\lambda_i \in \mathbb{R}$ .*

*Proof.* If  $\lambda_i \geq 0$ , choose a permutation coupling maximizing  $q_i$ . By the single-coordinate range, this gives  $q_i = 2p_i$ , so

$$\lambda_i q_i - \lambda_i / 2 = \lambda_i (2p_i - 1/2) \geq 0.$$

If  $\lambda_i < 0$ , choose the diagonal coupling, giving  $q_i = p_i$ . Then

$$\lambda_i q_i - \lambda_i / 2 = \lambda_i (p_i - 1/2) \geq 0,$$

because both factors are negative.  $\square$

The first genuinely mixed case is therefore a positive low-critical coordinate compensated by one or more negative coordinates. There is a clean matching condition that resolves many such directions.

Let  $i$  be the positive coordinate and let  $J$  be the set of negative coordinates. Partition  $\mathcal{F}$  by the trace on  $J$ : two sets lie in the same block if they contain exactly the same coordinates from  $J$ .

**Lemma 2** (Protected one-positive boosting). *Assume  $1/4 \leq p_i < 1/2$ . Suppose that for every  $J$ -trace block  $\mathcal{B} \subseteq \mathcal{F}$ ,*

$$|\{A \in \mathcal{B} : i \in A\}| \leq \frac{|\mathcal{B}|}{2}.$$

*Then the weighted critical assignment inequality holds for every signed weight vector with  $\lambda_i > 0$  and  $\lambda_j \leq 0$  for all  $j \in J$ .*

*Proof.* Inside each  $J$ -trace block  $\mathcal{B}$ , choose a bijection  $\sigma_{\mathcal{B}} : \mathcal{B} \rightarrow \mathcal{B}$  such that every column containing  $i$  is assigned to a row omitting  $i$ . This is possible because the number of  $i$ -columns in the block is at most the number of rows omitting  $i$  in the same block. Combining these bijections gives a permutation  $\sigma$  of  $\mathcal{F}$ .

Since  $\sigma$  preserves every  $J$ -trace block, it does not increase any negative coordinate:  $q_j(\sigma) = p_j$  for each  $j \in J$ . Since every  $i$ -column is assigned to an  $i$ -missing row, the extra contribution to  $q_i$  is  $p_i$ , so  $q_i(\sigma) = 2p_i$ . Hence

$$\sum_k \lambda_k q_k(\sigma) - \frac{1}{2} \sum_k \lambda_k = \lambda_i (2p_i - 1/2) + \sum_{j \in J} \lambda_j (p_j - 1/2) \geq 0.$$

The first term is nonnegative because  $p_i \geq 1/4$ , and each summand in the second term is nonnegative because  $\lambda_j \leq 0$  and  $p_j < 1/2$ .  $\square$

Thus any obstruction not handled by this lemma must have the positive coordinate too dense inside some trace block of the negative coordinates. This is a local obstruction, not a global entropy phenomenon.

There is a more flexible version that allows dense blocks as long as the negative weights leave enough slack.

**Lemma 3** (Trace-preserving partial boost). *Let  $i$  be the unique coordinate with positive weight, and let  $J$  be the set of coordinates with negative weights. Partition  $\mathcal{F}$  into blocks by the  $J$ -trace. For a block  $\mathcal{B}$ , write*

$$r_{\mathcal{B}} = |\{A \in \mathcal{B} : i \in A\}|, \quad \ell_{\mathcal{B}} = |\{A \in \mathcal{B} : i \notin A\}|.$$

Set

$$\Delta_{i|J} = \frac{1}{m} \sum_{\mathcal{B}} \min(r_{\mathcal{B}}, \ell_{\mathcal{B}}).$$

Then there is a permutation coupling such that

$$q_j = p_j \quad (j \in J), \quad q_i = p_i + \Delta_{i|J}.$$

Consequently the weighted critical assignment inequality holds whenever

$$\Delta_{i|J} \geq \left(\frac{1}{2} - p_i\right) + \frac{1}{\lambda_i} \sum_{j \in J} \lambda_j \left(\frac{1}{2} - p_j\right).$$

*Proof.* Inside each  $J$ -trace block  $\mathcal{B}$ , match as many  $i$ -containing columns as possible to  $i$ -omitting rows. This creates exactly  $\min(r_{\mathcal{B}}, \ell_{\mathcal{B}})$  new OR occurrences of  $i$  inside the block. Complete the partial matching arbitrarily to a bijection of  $\mathcal{B}$ . Combining these bijections over all blocks gives a permutation of  $\mathcal{F}$  preserving every  $J$ -trace, hence preserving  $q_j = p_j$  for all  $j \in J$ , and increasing  $q_i$  by  $\Delta_{i|J}$ .

For this coupling,

$$\sum_k \lambda_k (q_k - 1/2) = \lambda_i \left(p_i + \Delta_{i|J} - \frac{1}{2}\right) + \sum_{j \in J} \lambda_j \left(p_j - \frac{1}{2}\right).$$

The displayed lower bound on  $\Delta_{i|J}$  is exactly the condition that this quantity is nonnegative.  $\square$

The protected-block lemma is the special case where  $r_{\mathcal{B}} \leq \ell_{\mathcal{B}}$  in every block, so  $\Delta_{i|J} = p_i$ .

## 7.4 Multi-Positive Trace Blocks

Now let  $P = \{i : \lambda_i > 0\}$  and  $J = \{j : \lambda_j < 0\}$ . The natural generalization is to preserve the  $J$ -trace while optimizing the positive coordinates inside each trace block.

For a  $J$ -trace block  $\mathcal{B}$ , define the positive gain weight

$$g_{\lambda}^+(A, B) = \sum_{i \in P \setminus A} \lambda_i \mathbf{1}_{i \in B}.$$

Let

$$M_{\mathcal{B}}(\lambda) = \max_{\sigma_{\mathcal{B}} \in S_{\mathcal{B}}} \sum_{A \in \mathcal{B}} g_{\lambda}^+(A, \sigma_{\mathcal{B}}(A)).$$

This is an ordinary assignment problem inside the block.

**Lemma 4** (Trace-preserving positive reduction). *If*

$$\sum_{\mathcal{B}} M_{\mathcal{B}}(\lambda) \geq m \sum_{i \in P} \lambda_i \left( \frac{1}{2} - p_i \right) + m \sum_{j \in J} \lambda_j \left( \frac{1}{2} - p_j \right),$$

*then the weighted critical assignment inequality holds for  $\lambda$ .*

*Proof.* For each  $J$ -trace block  $\mathcal{B}$ , choose a maximizing permutation  $\sigma_{\mathcal{B}}$  in the definition of  $M_{\mathcal{B}}(\lambda)$ . Combining these permutations gives a permutation  $\sigma$  of  $\mathcal{F}$  preserving the  $J$ -trace of every set. Hence  $q_j(\sigma) = p_j$  for all  $j \in J$ . The positive gain of this permutation is exactly  $\sum_{\mathcal{B}} M_{\mathcal{B}}(\lambda)$ . Therefore

$$\sum_i \lambda_i (q_i(\sigma) - 1/2) = \frac{1}{m} \sum_{\mathcal{B}} M_{\mathcal{B}}(\lambda) - \sum_{i \in P} \lambda_i \left( \frac{1}{2} - p_i \right) - \sum_{j \in J} \lambda_j \left( \frac{1}{2} - p_j \right).$$

The assumed inequality says that the right-hand side is nonnegative, which is the weighted critical assignment inequality.  $\square$

A cheaper sufficient condition comes from random block permutations. If  $p_{i|\mathcal{B}}$  denotes the conditional frequency of coordinate  $i$  inside the block  $\mathcal{B}$ , then averaging a uniform random permutation of each block gives expected positive gain

$$\sum_{\mathcal{B}} |\mathcal{B}| \sum_{i \in P} \lambda_i p_{i|\mathcal{B}} (1 - p_{i|\mathcal{B}}).$$

Thus the trace-preserving reduction holds whenever

$$\frac{1}{m} \sum_{\mathcal{B}} |\mathcal{B}| \sum_{i \in P} \lambda_i p_{i|\mathcal{B}} (1 - p_{i|\mathcal{B}}) \geq \sum_{i \in P} \lambda_i \left( \frac{1}{2} - p_i \right) + \sum_{j \in J} \lambda_j \left( \frac{1}{2} - p_j \right).$$

**Problem 3** (Trace-block boost problem). *Determine when a remaining-cone weight vector satisfies the trace-preserving positive reduction. When it fails, quantify how much negative-coordinate “leakage” is needed to compensate for the missing trace-preserving boost.*

The trace-preserving positive reduction is not the final theorem. It is a useful sufficient condition, but it can fail even while the full assignment inequality succeeds.

**Example 1** (Controlled leakage is sometimes necessary). *There is a reduced union-closed family on six coordinates with  $|\mathcal{F}| = 27$  and two maximal critical coordinates  $i, j$  such that*

$$p_i = \frac{8}{27}, \quad p_j = \frac{11}{27}.$$

*Take weights*

$$\lambda_i = 1, \quad \lambda_j = -\frac{1}{25}.$$

*This weight vector lies in the remaining cone:*

$$\lambda_i \left( \frac{1}{2} - p_i \right) + \lambda_j \left( \frac{1}{2} - p_j \right) = \frac{1}{5} > 0,$$

*and*

$$\lambda_i \left( 2p_i - p_i^2 - \frac{1}{2} \right) + \lambda_j \left( 2p_j - p_j^2 - \frac{1}{2} \right) = -\frac{7}{6075} < 0.$$

If we preserve the  $j$ -trace, the best boost gives

$$q_i = \frac{13}{27}, \quad q_j = \frac{11}{27},$$

and the weighted slack is negative:

$$\lambda_i(q_i - 1/2) + \lambda_j(q_j - 1/2) = -\frac{2}{135} < 0.$$

However the unrestricted assignment can choose

$$q_i = \frac{16}{27}, \quad q_j = \frac{14}{27},$$

giving positive weighted slack

$$\lambda_i(q_i - 1/2) + \lambda_j(q_j - 1/2) = \frac{62}{675} > 0.$$

Thus the proof cannot insist on preserving every negative coordinate. It must allow controlled leakage in the negative coordinates when the positive boost gained is sufficiently larger.

**Problem 4** (Controlled-leakage assignment). *For a remaining-cone weight vector, prove that either the trace-preserving positive reduction succeeds, or there is an assignment whose additional positive gain exceeds the weighted cost of increasing the negative coordinates. The two-coordinate case should be attacked first, since the example above already shows the necessary phenomenon.*

## 7.5 Trace-Count Reduction

The controlled-leakage problem has an important simplification: for a fixed coordinate set, the OR-marginal polytope depends only on the counts of coordinate traces.

Let  $I$  be a finite set of coordinates. For a trace  $u \in \{0, 1\}^I$ , write

$$n_u = |\{A \in \mathcal{F} : A|_I = u\}|.$$

For a uniform coupling of two members of  $\mathcal{F}$ , let  $T_{u,v}$  be the amount of mass transported from trace  $u$  to trace  $v$ , scaled by  $m = |\mathcal{F}|$ . Then  $T$  is a nonnegative matrix satisfying

$$\sum_v T_{u,v} = n_u, \quad \sum_u T_{u,v} = n_v.$$

Conversely, every such transportation matrix is realized by a convex combination of uniform couplings of  $\mathcal{F}$ .

**Proposition 5** (Trace-count reduction). *For a fixed coordinate set  $I$ , the set of possible OR-marginal vectors  $q_I(\pi)$  over all uniform self-couplings  $\pi$  is exactly the image of the transportation polytope*

$$\{T_{u,v} \geq 0 : \sum_v T_{u,v} = n_u, \sum_u T_{u,v} = n_v\}$$

under the linear map

$$q_i(T) = \frac{1}{m} \sum_{u,v: (u \vee v)_i = 1} T_{u,v}.$$

*Proof.* Any coupling of  $\mathcal{F}$  pushes forward to such a transportation matrix on  $I$ -traces, and the displayed formula for  $q_i$  is immediate. Conversely, given  $T$ , distribute the mass  $T_{u,v}$  arbitrarily among the complete bipartite pairs between the sets with trace  $u$  and the sets with trace  $v$ . This gives a coupling of the uniform measure on  $\mathcal{F}$  with itself.  $\square$

Thus the two-coordinate controlled-leakage problem is a statement about a  $4 \times 4$  transportation polytope, independent of union-closure except through which trace-count vectors can occur in reduced maximal critical families.

**Proposition 6** (Two-coordinate critical centering). *Let  $I = \{i, j\}$ . If*

$$\frac{1}{4} \leq p_i < \frac{1}{2}, \quad \frac{1}{4} \leq p_j < \frac{1}{2},$$

*then the trace-count transportation polytope contains a coupling with*

$$q_i = q_j = \frac{1}{2}.$$

*Equivalently, every two-coordinate weighted critical assignment inequality holds, with no union-closure hypothesis needed beyond the realized trace counts.*

*Proof.* Normalize the trace counts by  $m = |\mathcal{F}|$ , and write

$$a = \mathbb{P}[00], \quad b = \mathbb{P}[10], \quad c = \mathbb{P}[01], \quad d = \mathbb{P}[11].$$

Thus

$$p_i = b + d, \quad p_j = c + d.$$

Start with the diagonal self-coupling. If we move an amount  $\varepsilon$  symmetrically from the two diagonal atoms  $u, u$  and  $v, v$  to the off-diagonal atoms  $u, v$  and  $v, u$ , then the two marginals are unchanged. This swap increases  $q_i$  by  $\varepsilon$  exactly when the traces  $u$  and  $v$  differ in coordinate  $i$ , and similarly for  $j$ .

It is therefore enough to choose nonnegative edge weights on the complete graph on the four traces, with incident weight at a trace at most its mass, so that the total weight crossing the  $i$ -cut is

$$D_i = \frac{1}{2} - p_i = \frac{1}{2} - b - d$$

and the total weight crossing the  $j$ -cut is

$$D_j = \frac{1}{2} - p_j = \frac{1}{2} - c - d.$$

The assumptions give  $0 \leq D_i, D_j \leq 1/4$ .

Assume first that  $c \geq b$ ; the other case follows by interchanging  $i$  and  $j$ . Then

$$D_i - D_j = c - b =: E.$$

We first spend this excess  $i$ -demand on edges which cross only the  $i$ -cut. Put

$$z = \min(d, E) \quad \text{on the edge } 01\text{--}11, \quad u = E - z \quad \text{on the edge } 00\text{--}10.$$

The choice of  $z$  is plainly feasible. If  $u > 0$ , then  $E > d$ , so  $u = c - b - d$ . The inequality  $u \leq b$  follows from

$$c \leq \frac{1}{2} - d \leq 2b + d,$$

where the last inequality is  $b + d \geq 1/4$ . Also  $u \leq a$  follows from  $c < 1/2$ . Thus the excess step is feasible. After it, the two remaining demands are both equal to  $D_j$ .

There are two cases. If  $E \leq d$ , then the residual masses at the traces 00, 10, 01, 11 are

$$a, \quad b, \quad b, \quad d - E.$$

Use  $\min(b, D_j)$  on the edge 10–01. If this does not exhaust the remaining demand, the leftover amount is

$$D_j - b = \frac{1}{2} - b - c - d = a - \frac{1}{2}.$$

This is at most  $a$ , and it is at most  $d - E = d - c + b$  because

$$\frac{1}{2} - b - c - d \leq d - c + b \iff \frac{1}{2} \leq 2(b + d).$$

The last inequality is again  $p_i \geq 1/4$ . Hence the leftover can be put on the edge 00–11.

If  $E > d$ , then after the excess step the trace 11 has no residual mass, and we put all of the remaining demand  $D_j$  on the edge 10–01. This fits at trace 10 because

$$D_j \leq 2b + d - c \iff \frac{1}{2} \leq 2(b + d),$$

and it fits at trace 01 because

$$D_j \leq c - d \iff c \geq \frac{1}{4}.$$

The latter follows from  $E > d$ , since then  $c > b + d = p_i \geq 1/4$ .

In all cases we have built feasible symmetric swaps whose  $i$ - and  $j$ -cut weights are exactly  $D_i$  and  $D_j$ . Applying these swaps to the diagonal coupling gives a self-coupling with

$$q_i = p_i + D_i = \frac{1}{2}, \quad q_j = p_j + D_j = \frac{1}{2}.$$

Scaling back by  $m$  gives the required point in the trace-count transportation polytope.  $\square$

The script still contains an exhaustive sanity check for this proposition on all two-coordinate trace count vectors with total size at most 16. The check is deliberately trace-count based: it enumerates  $4 \times 4$  transportation matrices rather than permutations of the full family. This makes the controlled-leakage phenomenon visible as a small transportation fact.

## 7.6 The First Higher-Dimensional Test

The proof just given has a useful general form. For a coordinate set  $I$ , let  $\mu_u$  be the normalized mass of trace  $u \in \{0, 1\}^I$ , and put

$$D_i = \frac{1}{2} - p_i.$$

Starting from the diagonal self-coupling, a symmetric swap of mass  $\varepsilon$  between two traces  $u$  and  $v$  preserves both marginals and increases  $q_i$  by  $\varepsilon$  exactly when  $u_i \neq v_i$ . Therefore the following finite certificate is enough to center all coordinates in  $I$ : find edge weights  $\varepsilon_{u,v} \geq 0$  on unordered pairs of traces such that

$$\sum_{v \neq u} \varepsilon_{u,v} \leq \mu_u \quad \text{for every trace } u$$

and

$$\sum_{u_i \neq v_i} \varepsilon_{u,v} = D_i \quad \text{for every } i \in I.$$

The two-coordinate proposition proves that this capacitated cut-demand problem is always feasible for  $|I| = 2$  when both coordinates are critical.

The naive higher-dimensional version is already false if one ignores union-closure. For three coordinates, take normalized trace counts with total mass 7:

$$n_{100} = n_{010} = n_{001} = 2, \quad n_{011} = 1,$$

and all other trace counts zero. Then

$$(p_1, p_2, p_3) = \left( \frac{2}{7}, \frac{3}{7}, \frac{3}{7} \right),$$

so all three coordinates are critical, but an exact enumeration of the  $8 \times 8$  trace transportation polytope finds no coupling with  $q_1 = q_2 = q_3 = 1/2$ . This is not a legal trace support for a union-closed family: for instance  $100 \vee 010 = 110$  is missing. Thus the third coordinate is precisely where the join-closure of the trace support has to become part of the proof, not just the one-coordinate marginals.

A small exhaustive check first seemed to support this diagnosis: among the 147 three-coordinate critical trace count vectors of total size at most 9 whose positive support is union-closed, the script found no miss. But the next layer produces a genuine obstruction.

**Proposition 7** (A parametric three-coordinate trace obstruction). *Let  $k \geq 2$ . For three coordinates  $x, y, z$ , take trace counts*

$$n_{000} = 4, \quad n_{110} = k, \quad n_{001} = k, \quad n_{011} = 2, \quad n_{111} = 1,$$

*and all other trace counts zero. The positive trace support is union-closed, and*

$$(p_x, p_y, p_z) = \left( \frac{k+1}{2k+7}, \frac{k+3}{2k+7}, \frac{k+3}{2k+7} \right),$$

*so all three coordinates are critical. Nevertheless the trace-count transportation polytope does not contain  $(1/2, 1/2, 1/2)$ .*

*Proof.* Use the signed functional

$$\varphi(q) = q_x - q_y - q_z.$$

At the desired center,  $\varphi = -1/2$ . On the support

$$S = \{000, 110, 001, 011, 111\},$$

define potentials

$$\tau(000) = 0, \quad \tau(110) = 0, \quad \tau(001) = -\frac{1}{2}, \quad \tau(011) = -1, \quad \tau(111) = 0.$$

A direct check of the  $5 \times 5$  table gives

$$\varphi(u \vee v) \leq \tau(u) + \tau(v) \quad \text{for all } u, v \in S.$$

Therefore every self-coupling of the trace distribution satisfies

$$\mathbb{E}[\varphi(U \vee V)] \leq 2 \mathbb{E}[\tau(U)] = 2 \cdot \frac{k(-1/2) + 2(-1)}{2k+7} = -\frac{k+4}{2k+7} < -\frac{1}{2}.$$

Thus the center is separated by the assignment-dual direction  $(1, -1, -1)$ . □

This obstruction is not merely an artificial non-realizable count vector. It is the projection, onto coordinates  $(1, 2, 4)$ , of the union-closed family on five coordinates

$$\{0, 1, 7, 8, 9, 15, 24, 25, 28, 29, 31\},$$

where sets are written as bit masks. But that lift has hidden Frankl witnesses: the two auxiliary coordinates have frequencies  $7/11$  and  $8/11$ . So the failed trace-count theorem is not a dead end; it is a sharper version of the entropy–Hall dichotomy. The three-coordinate dual obstruction can exist at the projected trace level, but in this lift the obstruction is paid for by heavy coordinates inside the fibers.

In fact this is forced for every lift of this obstruction profile.

**Lemma 5** (Bottom-fiber propagation). *Let  $\mathcal{F}$  be a union-closed family, projected onto a coordinate set  $I$ . Let  $A$  be the fiber over the zero trace  $0 \in \{0, 1\}^I$ , viewed as a family of auxiliary sets in the remaining coordinates. Suppose an auxiliary coordinate  $h$  appears in  $r$  members of  $A$ .*

*If  $u$  is any projected trace such that the fiber  $F_u$  has a maximum auxiliary set  $M_u$ , then  $h \in M_u$ . Consequently, if  $U$  is a collection of such traces with nonempty fibers, then  $h$  appears in at least*

$$r + |U|$$

*members of the original family, provided  $U$  does not include the zero trace.*

*Proof.* Choose a member  $a \in A$  containing  $h$ . Since the projected trace of  $a$  is zero, the union of  $a$  with any member of  $F_u$  still has projected trace  $u$ . In particular,  $a \cup M_u$  belongs to  $F_u$ . But  $M_u$  is maximum in that fiber and  $M_u \subseteq a \cup M_u$ , so  $a \cup M_u = M_u$ . Hence  $h \in M_u$ .  $\square$

**Lemma 6** (Four-set local Frankl). *Every four-element union-closed family of auxiliary sets has an element which appears in at least two members.*

*Proof.* If every member were empty there would not be four distinct sets. Suppose, to the contrary, that every auxiliary element appears in at most one member. Then any two nonempty members are disjoint. But the union of two distinct nonempty members contains every element of both, so each of those elements appears again in the union member. This is impossible. Therefore there cannot be two distinct nonempty members; with the empty set, this gives at most two distinct members total, again impossible. Hence some element appears in at least two members.  $\square$

**Lemma 7** (Small local Frankl bounds). *The following local bounds hold for finite union-closed families of auxiliary sets:*

$$\frac{|G|}{\max_h |\{g \in G : h \in g\}|} \begin{array}{c|cccc} & 4 & 5 & 6 & 7 \\ \hline & \geq 2 & \geq 3 & \geq 4 & \geq 4. \end{array}$$

*The six-set bound is deliberately stronger than the ordinary half bound.*

*Proof.* For  $|G| = 4$ , the preceding four-set local Frankl lemma proves the claim. For the other three cases we use a finite incidence-pattern reduction, which is also the intended Lean proof route.

Assume  $G$  has  $m$  members and no element reaches the claimed threshold  $t$ . Let  $M$  be the maximum member of  $G$ , and label its row by  $m - 1$ . Every active auxiliary coordinate then has an incidence column containing row  $m - 1$  and at most  $t - 2$  of the other rows. There are only

$$\sum_{j=0}^{t-2} \binom{m-1}{j}$$

such columns. The script enumerates every subset of these columns, keeps the ones that separate the  $m$  rows, and checks whether the resulting row set is closed under coordinatewise OR. For

$$(m, t) = (5, 3), (6, 4), (7, 4)$$

there is no such row set. This is a finite certificate, printed by `--local-frankl-checks`.  $\square$

**Lemma 8** (Horn propagation certificate). *Fix an auxiliary coordinate  $h$  obtained from the zero fiber. For each trace  $u$ , let  $y_u$  be the number of members in the  $u$ -fiber containing  $h$ , and let  $n_u$  be the fiber size. Then the following constraints are sound:*

$$y_0 \geq r_0, \quad y_u \geq 1 \quad (u \neq 0, n_u > 0),$$

where  $r_0$  is any valid local lower bound in the zero fiber, and

$$y_{u \vee v} = n_{u \vee v} \implies y_u = n_u \text{ or } y_v = n_v.$$

*Proof.* The first inequality is the chosen local zero-fiber bound. The second is bottom-fiber propagation: each nonzero fiber maximum contains  $h$ .

For the Horn implication, suppose all members of the  $(u \vee v)$ -fiber contain  $h$ . If some member of the  $u$ -fiber omitted  $h$  and some member of the  $v$ -fiber also omitted  $h$ , then their union would omit  $h$  but would lie in the  $(u \vee v)$ -fiber, a contradiction. Hence one of the two input fibers must consist entirely of members containing  $h$ .  $\square$

**Proposition 8** (The parametric obstruction forces a hidden witness). *Let  $k \geq 2$ . Let  $\mathcal{F}$  be a union-closed family whose projection onto three coordinates has trace counts*

$$n_{000} = 4, \quad n_{110} = k, \quad n_{001} = k, \quad n_{011} = 2, \quad n_{111} = 1,$$

*with all other trace counts zero. If all members with the same trace are viewed as auxiliary sets in the remaining coordinates, then some auxiliary coordinate belongs to at least  $k + 5$  of the  $2k + 7$  members of  $\mathcal{F}$ . In particular, it is a Frankl witness.*

*Proof.* Let  $A$  be the 000 fiber. Since  $000 \vee 000 = 000$ , the four auxiliary sets in  $A$  form a union-closed family. By the four-set local Frankl lemma, this fiber has an element  $h$  which appears in at least two members of  $A$ .

Let  $B, C, D, T$  be the 110, 001, 011, and 111 fibers respectively. Every finite union-closed fiber has a maximum auxiliary set, so bottom-fiber propagation already shows that  $h$  appears in the maxima of all four nonzero fibers. If  $h$  appears in every member of  $B$ , then  $h$  appears in at least

$$2 + k + 1 + 1 + 1 = k + 5$$

members of the original family.

Otherwise choose  $b \in B$  with  $h \notin b$ . Since  $110 \vee 001 = 111$ , for every  $c \in C$  the union  $b \cup c$  is the unique member of the 111 fiber; therefore  $h \in c$ . Similarly, since  $110 \vee 011 = 111$ , every member of  $D$  contains  $h$ . Together with one occurrence in the maximum of  $B$ , this gives at least

$$2 + 1 + k + 2 + 1 = k + 6$$

occurrences. In either case  $h$  appears at least  $k + 5$  times, while

$$\frac{|\mathcal{F}|}{2} = \frac{2k + 7}{2} < k + 5.$$

$\square$

**Corollary 1** (The first sign obstructions are fiber-paid). *For three-coordinate critical trace counts of total size at most 16, the script’s propagation scan*

`--trace-three-propagation-scan-size 16`

*finds that every  $\{-1, 0, 1\}$ -separated class is paid for by the hidden element propagation certificate above.*

*Proof.* This is a finite computation. For each separated trace-count vector, the script minimizes  $\sum_u y_u$  subject to the small local zero-fiber bounds, bottom-fiber propagation, and the Horn implications above. In every case of total size at most 16, the minimum is at least half the total family size.  $\square$

At total size 17, the same scan reports the first unpaid classes: 66 sign-separated trace counts, of which 48 are paid by propagation and four canonical classes remain unpaid. Three of these have zero-fiber size 8, where the local zero-fiber bound has not yet been added. The remaining class

$$(6, 0, 0, 3, 3, 0, 3, 2)$$

has propagation lower bound 8, exactly one occurrence short of half. This is the current next obstruction layer.

The refined target is therefore not a pure trace-count theorem. It is a fiber-sensitive statement: if a projected three-coordinate obstruction of this kind occurs in an actual union-closed family, then either controlled transport using the internal fibers restores centering, or the same dual certificate can be lifted into a genuine Frankl witness outside the projected coordinates. The command `--trace-three-obstruction` prints this trace obstruction, its separating direction, and the explicit lift above.

## 7.7 Lean-Ready Boundary

The part of the argument that now looks stable enough to formalize is not the whole Frankl strategy, but the first obstruction layer. It breaks into a small reusable API:

1. A finite nonempty union-closed family has a maximum member, namely the union of all its members.
2. Bottom-fiber propagation: if a zero-trace auxiliary coordinate appears in a bottom fiber, then it appears in the maximum member of every nonzero fiber.
3. Small local Frankl bounds for zero fibers of sizes 4, 5, 6, and 7.
4. The Horn propagation certificate for fibers whose joins land in all- $h$  fibers.
5. The parametric three-coordinate trace obstruction is separated by the explicit assignment-dual potential above.
6. Every union-closed lift of that parametric obstruction has a hidden Frankl witness.

These are local, finite, and do not depend on any unproved global form of Frankl’s conjecture. They are therefore good candidates for the next Lean pass before attempting to formalize the broader entropy–transport program.

## 7.8 Coordinate-Order Faces

Some tight dual directions are structural and should be stripped away before looking for a real obstruction. Define a preorder on coordinates by

$$i \preceq j \iff \forall A \in \mathcal{F}, i \in A \Rightarrow j \in A.$$

Equivalently,  $\chi_i \leq \chi_j$  pointwise. If  $i \preceq j$ , then every coupled OR marginal satisfies

$$q_i(\pi) \leq q_j(\pi).$$

Indeed,  $i \in A \cup B$  implies  $i \in A$  or  $i \in B$ , hence  $j \in A$  or  $j \in B$ , so  $j \in A \cup B$ .

Thus the hyperplane  $q_i = q_j$  can support the centering polytope at  $(1/2, \dots, 1/2)$  without representing a failure. The corresponding signed dual direction is  $\lambda_i = 1, \lambda_j = -1$ . Clone coordinates give the special case  $i \preceq j$  and  $j \preceq i$ , where  $q_i = q_j$  for all couplings.

This suggests the following normalization before attacking the assignment dual:

quotient coordinate clones, then keep only maximal coordinates under  $\preceq$ .

For Frankl this is harmless as a search principle: if  $i \preceq j$ , then  $p_i \leq p_j$ , so the dominated coordinate  $i$  is never a better witness than  $j$ . In a putative counterexample all maximal coordinates are still light.

**Problem 5** (Face-stripped assignment inequality). *Prove the weighted critical assignment inequality after quotienting clone coordinates and restricting to maximal critical coordinates. Then solve the extension problem for dominated coordinates with compatible targets  $q_i \leq q_j$  when  $i \preceq j$ . Exact centering of every dominated coordinate may be more than the entropy argument needs; a target inside  $p_i < q_i < 1 - p_i$  would still give coordinate entropy gain.*

## 8 The First Novel Program

**Problem 6** (Optimal coupled OR). *Among all couplings  $\pi \in \Pi(\mathcal{F})$ , maximize*

$$\Phi(\pi) = H(X \vee Y) - H(X).$$

*Understand the dual certificate when the maximum is nonpositive.*

This problem is nonlinear because of entropy, but it has a transport flavor: we are choosing mass on  $\mathcal{F} \times \mathcal{F}$  with fixed marginals, and the objective depends only on the OR fibers

$$m^{-1}(C) = \{(A, B) : A \vee B = C\}.$$

The failure of entropy gain should therefore be witnessed by a potential on OR fibers and coordinates.

**Heuristic 1** (Entropy failure equals collision structure). *If no coupling gives positive entropy advantage, then the OR map*

$$m : \mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F}, \quad m(A, B) = A \vee B,$$

*has enough forced collision/dependence structure to produce a combinatorial certificate. That certificate should be interpretable as a Hall obstruction to naive pairing.*

## 9 Hall Certificates for Pairing Failure

Fix a coordinate  $i$ . Let

$$L_i = \{A \in \mathcal{F} : i \notin A\}, \quad R_i = \{B \in \mathcal{F} : i \in B\}.$$

Frankl for coordinate  $i$  is exactly

$$|L_i| \leq |R_i|.$$

Build a bipartite graph  $G_i$  from  $L_i$  to  $R_i$  by declaring that  $A \in L_i$  is adjacent to  $B \in R_i$  if

$$B = A \cup C \quad \text{for some } C \in \mathcal{F} \text{ with } i \in C.$$

Union-closure guarantees  $B \in \mathcal{F}$  whenever  $C$  is chosen.

**Lemma 9** (Matching certificate, desired). *If  $G_i$  has a matching saturating  $L_i$ , then  $i$  is a Frankl element.*

This is immediate from Hall's theorem. The point is what happens when no such matching exists.

**Definition 2** (Hall defect). *For  $U \subseteq L_i$ , define*

$$\delta_i(U) = |U| - |N_i(U)|.$$

*A positive Hall defect is a certificate that the obvious union-pairing strategy for coordinate  $i$  cannot work.*

In a counterexample, every coordinate has  $|L_i| > |R_i|$ , so every coordinate has global counting pressure against Frankl. But the graphs  $G_i$  are not independent: all are induced by the same join operation on  $\mathcal{F}$ .

**Problem 7** (Incompatibility of Hall defects). *Show that a finite union-closed family cannot support coherent positive Hall defects for every active coordinate, unless some coordinate has  $p_i \geq 1/2$ .*

## 10 The Entropy–Hall Synthesis

The central proposal is a dichotomy.

**Conjecture 2** (Entropy–Hall dichotomy). *Let  $\mathcal{F}$  be a finite union-closed family with  $0 < p_i < 1/2$  for every active coordinate. Then at least one of the following holds:*

- (a) *There exists a coupling  $\pi \in \Pi(\mathcal{F})$  with  $\Phi(\pi) > 0$ .*
- (b) *The dual obstruction to  $\Phi(\pi) > 0$  yields Hall-defect data  $\{\delta_i(U_i) > 0\}_i$  that is incompatible with union-closure.*

The first alternative gives an entropy contradiction. The second alternative is meant to convert the entropy method's slack into a purely finite combinatorial contradiction.

## 11 Encountering the Barriers One by One

### 11.1 Barrier: iid OR stops around $(3 - \sqrt{5})/2$

Path through: replace iid OR by a positively correlated coupling. The target is not arbitrary dependence, but dependence tuned so that

$$p_i < q_i(\pi) < 1 - p_i$$

for all coordinates. This preserves positive coordinate entropy gain while still moving away from the original distribution.

Concrete test:

Can every putative counterexample admit such a coupling?

If not, what is the transport dual obstruction?

### 11.2 Barrier: entropy gains are eaten by dependence

Path through: measure the loss explicitly as

$$\text{TC}(Z) - \text{TC}(X).$$

This loss is not noise; it records collision structure of the OR map. Try to show that large dependence loss implies a strong Hall-defect pattern.

Concrete target:

$$\text{TC}(Z) - \text{TC}(X) \leq \text{a weighted sum of local collision defects.}$$

If the right side is too large, extract a coordinate with an actual matching or fractional matching.

### 11.3 Barrier: average-size bounds lose dimension

Path through: introduce an effective dimension for the separating coordinate system

$$\{\chi_i : \mathcal{F} \rightarrow \mathbf{2}\}.$$

Possible choices:

- (i) rank of the incidence matrix over  $\mathbb{Q}$ ;
- (ii) entropy dimension,  $\sum_i h(p_i)$  normalized by  $\log |\mathcal{F}|$ ;
- (iii) number of join-irreducible coordinates after quotienting clones;
- (iv) VC or shatter dimension of the coordinate traces.

Then prove a trichotomy:

$$\text{dense} \quad \vee \quad \text{low effective dimension} \quad \vee \quad \text{entropy-transport contradiction.}$$

### 11.4 Barrier: dense cases are not the hard cases

Path through: use known density theorems as a boundary condition, but focus the new argument on sparse families. Sparse families have larger OR-collision fibers, which is exactly the regime where the entropy/Hall synthesis should have more structure to exploit.

## 11.5 Barrier: equivalent formulations move the problem

Path through: choose the formulation by the obstruction it exposes.

|                   |                    |                                  |
|-------------------|--------------------|----------------------------------|
| set families      | $\rightsquigarrow$ | frequency and pairing,           |
| join-semilattices | $\rightsquigarrow$ | coordinate homomorphisms,        |
| graphs            | $\rightsquigarrow$ | maximal stable set counts,       |
| closure systems   | $\rightsquigarrow$ | Galois duality and irreducibles. |

The proposed note lives in the join-semilattice/probability hybrid because it keeps both the OR operation and the counting measure visible.

## 12 Concrete Lemmas to Try

**Lemma 10** (Coupled OR window). *Assume  $\max_i p_i < 1/2$ . Either there exists a coupling  $\pi \in \Pi(\mathcal{F})$  such that*

$$p_i < q_i(\pi) < 1 - p_i \quad \text{for all active } i,$$

*or the transport dual yields a nonzero signed measure on coordinates that separates the family from this window.*

**Lemma 11** (Window plus bounded correlation loss). *If a coupling  $\pi$  satisfies the OR window and*

$$\text{TC}(X \vee Y) - \text{TC}(X) < \sum_i (h(q_i(\pi)) - h(p_i)),$$

*then  $\mathcal{F}$  satisfies Frankl.*

This lemma is tautological, but useful: it isolates the only remaining obstacle as correlation growth.

**Lemma 12** (Correlation loss gives Hall defect). *For an entropy-maximizing coupling  $\pi$ , large positive  $\text{TC}(X \vee Y) - \text{TC}(X)$  forces a positive weighted combination of Hall defects in the graphs  $G_i$ .*

This is the first genuinely speculative lemma. It asks us to turn dependence created by OR-collisions into a finite matching obstruction.

**Lemma 13** (Hall defects uncross). *For each coordinate  $i$ , among all positive Hall defects choose an inclusion minimal  $U_i \subseteq L_i$ . The family of minimal defects can be chosen to satisfy an uncrossing property compatible with joins:*

$$U_i \vee U_j = \{A \vee B : A \in U_i, B \in U_j\}$$

*has controlled neighborhood growth.*

**Lemma 14** (No coherent defect web). *No reduced finite join-semilattice with separating coordinates and  $p_i < 1/2$  for all  $i$  admits a coherent system of uncrossed positive Hall defects for every coordinate.*

If the last two lemmas are true in an appropriately sharpened form, they would complete the entropy–Hall dichotomy.

## 13 A Minimal-Counterexample Normalization

Suppose  $\mathcal{F}$  is a counterexample of minimal size. We should be able to enforce the following normalizations:

- (N1) The ground set is exactly  $\bigcup \mathcal{F}$ .
- (N2) Distinct coordinates induce distinct homomorphisms  $\chi_i : \mathcal{F} \rightarrow \mathbf{2}$ .
- (N3) No singleton belongs to  $\mathcal{F}$ , by the already formalized singleton case.
- (N4) Every coordinate has  $0 < p_i < 1/2$ .
- (N5) The family is closed under deleting irrelevant duplicate coordinates.

Each normalization should be formalized separately. If any normalization fails, it may already yield a smaller counterexample or a Frankl element.

## 14 Formalization Roadmap

The Lean project currently proves the singleton-containing case. The next formal targets should be low-risk statements that clarify the attack:

1. Define the split families  $L_i$  and  $R_i$  and prove  $i$  is Frankl iff  $|L_i| \leq |R_i|$ .
2. Define the bipartite graph  $G_i$  and prove the matching certificate using a finite Hall theorem from Mathlib, or first prove the fractional version via finite sums.
3. Define coordinate maps  $\chi_i$  as join-homomorphisms to ‘Bool’ or ‘Fin 2’, and prove they separate points after reducing the ground set.
4. Formalize the entropy identity

$$H(Z) - H(X) = \sum_i (h(q_i) - h(p_i)) - (\text{TC}(Z) - \text{TC}(X))$$

in a separate finite-probability file, likely before connecting it to Frankl.

5. Experiment computationally with small union-closed families, solving the coupled OR window as a linear feasibility problem.

## 15 What Would Count as Progress?

We do not need the full dichotomy immediately. Any of the following would be meaningful progress:

1. Find a natural class of sparse families where the coupled OR window exists and beats correlation growth.
2. Derive an explicit dual certificate for failure of the OR window.
3. Show that the dual certificate implies one of the known structural special cases.
4. Prove that minimal counterexamples must have a strong collision profile under the OR map.
5. Connect OR-collision fibers to Hall defects in one nontrivial family class.

## 16 First Computational Probe

The companion script

`window_experiment.py`

tests the coupled-OR window on small union-closed families. For exact small families it uses the Birkhoff-von Neumann theorem: every uniform coupling of two uniform samples from  $\mathcal{F}$  is a convex combination of permutation couplings. The script enumerates the corresponding OR-marginal vectors

$$q(\sigma)_i = \frac{1}{|\mathcal{F}|} |\{A \in \mathcal{F} : i \in A \cup \sigma(A)\}|$$

and solves the low-dimensional convex-hull LP

$$\max \gamma \quad \text{subject to} \quad p_i + \gamma \leq q_i \leq 1 - p_i - \gamma.$$

Two coordinate modes are useful:

1. **nonheavy**: coordinates with  $p_i \leq 1/2$ ;
2. **light**: coordinates with  $p_i < 1/2$ .

The first mode exposes exact boundary obstructions. The second mode tests whether strictly light coordinates can be pushed into the entropy window.

For all union-closed families on three ground elements, the exact **light** scan found no obstruction: every tested light-coordinate window had positive margin. The smallest margin occurred at  $p = 3/7$ , where the solver found  $q = 1/2$  and  $\gamma = 1/14$ . For rarer coordinates the optimum is often  $q = 2p$  rather than  $q = 1/2$ , exactly as predicted by the single-coordinate range. In **nonheavy** mode, the smallest margins are exactly zero and occur at the expected sharp boundary  $p_i = 1/2$ , including the Boolean cube itself.

For all union-closed families on four ground elements, with exact permutation couplings for  $|\mathcal{F}| \leq 8$  and sampled permutation couplings for larger families, the same pattern persisted. In a critical-coordinate scan ( $1/4 \leq p_i < 1/2$ ), all 1988 exact checks were center-feasible, and the sampled larger cases also found centers. The hardest light-coordinate examples had  $p = 7/15$  and sampled couplings reaching  $q = 1/2$ , giving  $\gamma = 1/30$ .

The script now also has a stronger **--scan-subsets** mode: instead of testing only the highest-frequency eligible coordinates, it tests every eligible coordinate subset of size at most three. In the exhaustive four-coordinate critical scan, this produced 4468 center checks:

$$2924 \text{ exact-feasible}, \quad 1544 \text{ sampled-feasible}, \quad 0 \text{ center misses}.$$

This is better evidence for the “obstruction-critical subcollection” phrasing of the conjecture than the earlier top-coordinate scan. Two small random subset scans also found no misses: 56 critical subset checks from 40 random five-coordinate generated families, and 37 critical subset checks from 30 random six-coordinate generated families.

The direct signed-weight stress test reinforces the face-stripping picture. With **--lambda-stress**, the script samples signed vectors  $\lambda$  and solves the exact assignment problem for each one. It now also reports the two necessary cone quantities from the diagonal and random-permutation baselines, so small raw slack is not mistaken for a genuine candidate obstruction. On the exhaustive four-coordinate critical subset scan, no sampled direction separated the center. Before normalization, many zero-slack directions were explained by clone coordinates or by coordinate-order faces  $i \preceq j$ , where  $q_i \leq q_j$  is forced for every coupling. After imposing both **--reduced-only** and

--maximal-only, the exhaustive four-coordinate scan had 628 center checks and no misses; the hardest sampled dual slack was

$$\frac{1}{30},$$

coming from a single maximal coordinate with  $p = 7/15$ . A reduced, maximal random six-coordinate scan produced only three maximal critical coordinate sets in the sample, again all single-coordinate, with hardest slack  $1/38$ . A broader random seven-coordinate scan produced 31 reduced/maximal critical checks, including a two-coordinate maximal critical case, again with no misses and no sampled separating direction; the hardest slack was  $1/70$ . This suggests that a large part of the apparent signed-dual difficulty is caused by forced faces rather than genuine transport obstructions.

A targeted search for remaining-cone geometry found mixed two-coordinate examples. One reduced six-coordinate family has maximal critical coordinates with frequencies

$$(p_i, p_j) = \left( \frac{8}{29}, \frac{14}{29} \right).$$

The best cone direction found had one positive low-critical weight on  $i$  and one negative high-critical weight on  $j$ . The exact assignment still had positive slack, and the maximizing permutation achieved

$$q_i = \frac{16}{29} = 2p_i, \quad q_j = \frac{14}{29} = p_j.$$

In other words, it boosted the rare positive coordinate all the way to its single-coordinate maximum while preserving the negative coordinate at the diagonal value. The protected-block condition explains this: inside the two  $j$ -trace blocks, the conditional frequencies of  $i$  were  $2/15$  and  $6/14$ , both at most  $1/2$ .

The stronger partial-boost lemma is needed because the protected-block condition is not automatic. A later search found a one-positive cone direction with

$$(p_j, p_i) = \left( \frac{19}{39}, \frac{17}{39} \right),$$

where  $j$  has negative weight and  $i$  has positive weight. In the  $j = 1$  trace block, the conditional frequency of  $i$  was  $13/19 > 1/2$ , so full boosting while preserving  $j$  was impossible. Nevertheless the trace-preserving boost size was

$$\Delta_{i|j} = \frac{10}{39},$$

and the exact assignment achieved

$$q_j = \frac{19}{39} = p_j, \quad q_i = \frac{27}{39} = p_i + \Delta_{i|j}.$$

For the tested cone direction the required boost was only about 0.0033, so the partial boost had large slack. This suggests that the right local quantity is not whether every block has conditional frequency at most  $1/2$ , but whether the aggregate trace-preserving boost  $\Delta_{i|J}$  exceeds the signed demand imposed by the negative coordinates.

A broader eight-coordinate reduced/maximal scan produced four remaining-cone candidate directions. All four were handled by the trace-preserving positive assignment. In the hardest one, the critical frequencies were

$$(p_i, p_j) = \left( \frac{21}{47}, \frac{23}{47} \right),$$

with positive weight on  $i$  and negative weight on  $j$ . The trace-preserving assignment kept

$$q_j = p_j$$

and boosted

$$q_i = \frac{42}{47} = 2p_i.$$

The required boost was only about 0.005, while the available trace-preserving boost was 21/47. This shows why the trace-preserving criterion remains valuable: many remaining-cone directions have tiny diagonal slack, but the available positive boost can still be enormous. The later controlled-leakage example shows that this criterion is not universal.

For random generated families on six ground elements, the sampled search again found positive window margins on the most critical light coordinates. Exact centering was usually found as the sampled permutation hull became denser, but some sampled “not seen feasible” cases remained. These are not certificates of failure; they are candidate stress tests for either a genuine dual obstruction or a more systematic sampling/LP method. The script can print the first such families with `--show-misses`.

For a sampled center miss, the script extracts a separating coordinate weight vector  $\lambda$  from the sampled  $q$ -hull. It then checks this separator against the full permutation-coupling polytope by solving the exact assignment problem

$$\max_{\sigma \in S_{\mathcal{F}}} \frac{1}{|\mathcal{F}|} \sum_{A \in \mathcal{F}} \sum_{i \in I} \lambda_i \mathbf{1}_{i \in A \cup \sigma(A)}.$$

If this exact maximum is still below

$$\sum_{i \in I} \lambda_i / 2,$$

then the separator is a genuine obstruction to simultaneous centering for the chosen coordinate set. In the first printed six-coordinate stress cases, the assignment check beat the sampled separator, so those misses were sampling weakness rather than real obstructions. This is encouraging: it suggests that many apparent misses are caused by a thin sampled hull, while also giving a clear route for recognizing a true obstruction if one appears.

The script also implements a column-generation version of this idea. Starting from sampled permutation couplings, it repeats:

- (1) if  $(1/2, \dots, 1/2)$  is in the current hull, stop;
- (2) otherwise extract a separating  $\lambda$ ;
- (3) solve the exact assignment problem for  $\lambda$ ;
- (4) if the exact optimum still separates, certify obstruction;
- (5) otherwise add the maximizing permutation as a new column.

On a stress run of 120 random six-coordinate generated families, using only 300 sampled permutations as the initial hull, all sampled critical-center misses were resolved by column generation. The resulting status counts were

3 exact-feasible, 77 sampled-feasible, 10 generated-feasible, 0 verified obstructions.

In the hardest printed case, the center was certified after adding just one exact assignment column. This strengthens the experimental evidence for critical-coordinate centering and gives a concrete finite procedure for searching for genuine obstructions.

These computations do not prove the coupled window lemma. They do suggest a sharper experimental conjecture:

**Conjecture 3** (Critical light-coordinate centering). *Let  $\mathcal{F}$  be finite and union-closed. For every set of light coordinates in the critical range*

$$I = \{i : 1/4 \leq p_i < 1/2\},$$

*or at least for every obstruction-critical subcollection of them, there exists a uniform coupling of  $X, Y$  such that*

$$\mathbb{P}[(X \vee Y)_i = 1] = \frac{1}{2} \quad \text{for all } i \in I.$$

For lighter coordinates  $p_i < 1/4$ , the right target is not  $1/2$  but some value in the interval

$$p_i < q_i \leq 2p_i < 1/2,$$

which still gives positive coordinate entropy gain. If the critical centering conjecture is true, then the iid entropy barrier is removed exactly where iid OR fails. The remaining obstruction would be entirely the total-correlation term

$$\text{TC}(X \vee Y) - \text{TC}(X),$$

which strengthens the case for connecting entropy loss with OR-collision and Hall-defect structure.

## 17 References

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